EXP 6 – QR BASED PARKING SYSTEM

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Introduction

QR Based Parking System is a smart parking solution where users can **scan a QR code to enter/exit parking**, check slot availability, and make payments digitally.
It reduces manual work, saves time, and improves parking efficiency.

Key Characteristics of RAD

• Focus on prototype development
• Continuous user feedback
• Fast development and iteration
• Reusable UI components
• Suitable for mobile/web applications

Phase 1: Requirements Planning

This phase identifies system requirements and user needs.

Features:

Navigation:
• Home
• Parking Slots
• QR Scan
• Payment
• Booking History

User Actions:
• View available parking slots
• Scan QR code to enter parking
• Book slot
• Make payment
• Exit using QR
• View history

Requirements:

1. Display available parking slots
2. Allow users to scan QR code
3. Show booking details
4. Enable online payment
5. Track entry and exit time
6. Generate confirmation message

Phase 2: User Design

Wireframes and UI prototypes are created.

Tools Used:

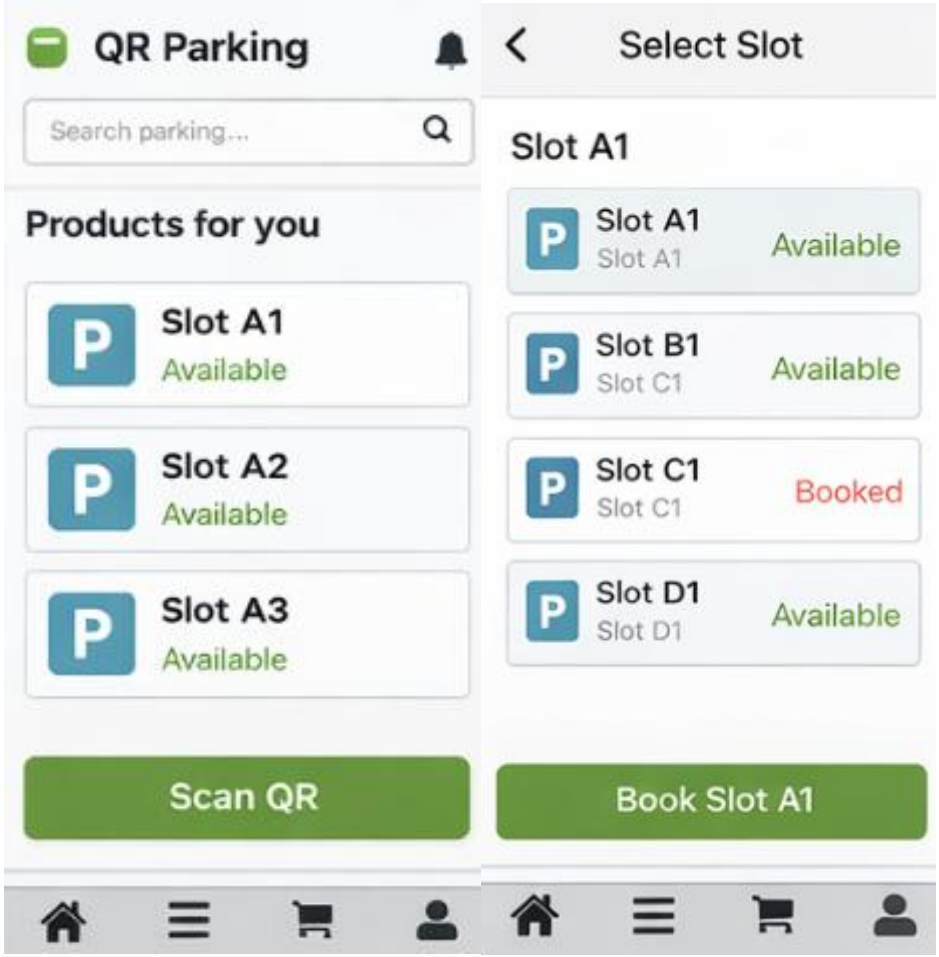
• Figma / Canva

Designed Screens:

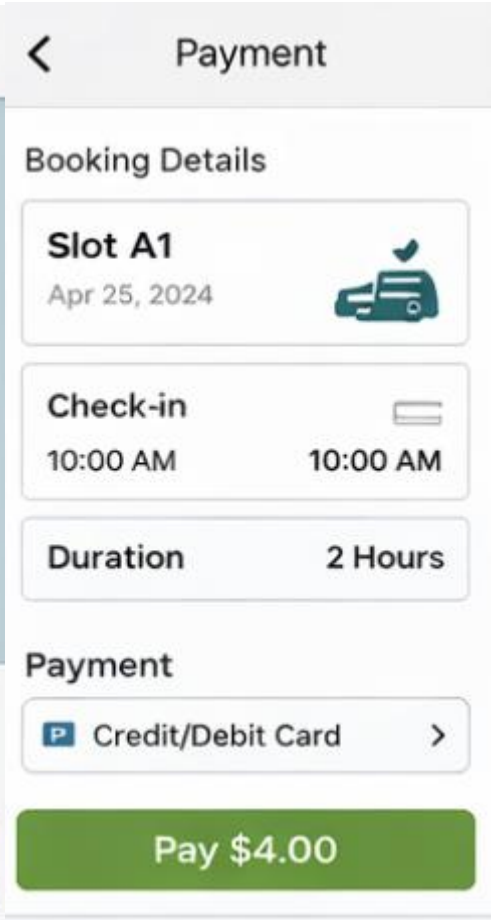
• Home Page – Shows parking availability
• Slot Selection Page – Choose parking slot
• QR Scanner Page – Scan entry QR
• Booking Confirmation Page
• Payment Page
• Exit Screen (QR scan)
• History Page

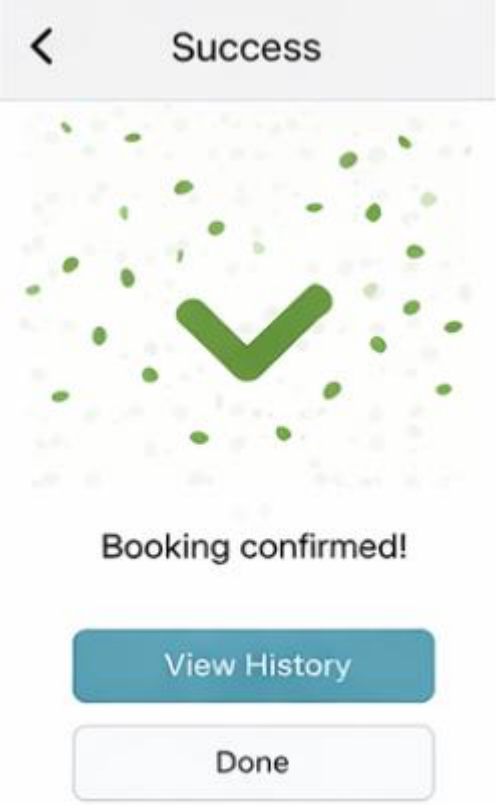
Design Features:

• Simple and user-friendly interface
• Clear navigation between screens
• Reusable components like header/footer
• Buttons for scan, book, and pay



Scan QR Code
Scan the QR code to enter parking area.
Scan QR


Payment
Booking Details
Slot A1
Apr 25, 2024
Check-in
10:00 AM
10:00 AM
Duration
2 Hours
Payment
Credit/Debit Card
Pay $4.00


Success
Booking confirmed!
View History
Done

Phase 3: Construction

Development and testing happen simultaneously.

Prototype Flow:

Home → Slot Selection → QR Scan → Booking → Payment → Confirmation → Exit

Dynamic Features:

• QR scanning simulation
• Slot availability updates
• Payment processing screen

Testing:

• Prototype tested using preview mode
• Feedback collected from users
• UI improved based on suggestions

Phase 4: Cutover

Final stage where system is ready for deployment.

Finalization:

• Prototype shared via Figma link
• Ready for development
• Can be converted to real app

User Support:

• User instructions provided
• Easy navigation for all users
• Documentation shared

Conclusion

The QR Based Parking System improves parking management by **reducing time, avoiding congestion, and enabling digital transactions**, making it efficient and user-friendly.